

## 1. Momentum

Momentum mainly tries to smooth the direction of the update.

A common formulation is:

$$v_t = \beta v_{t-1} + (1 - \beta) dW_t$$

Then:

$$W_t = W_{t-1} - \alpha v_t$$

Here,  $v_t$  represents a smoothed version of the gradient.

So the learning rate is multiplied directly by the velocity:

$$\Delta W = -\alpha v_t$$

The idea is:

"Based on the recent history of gradients, which direction should I move?"

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## 2. RMSProp

RMSProp has a different goal.

It wants to adapt the learning rate for each parameter based on the magnitude of its gradients.

It calculates:

$$s_t = \beta s_{t-1} + (1 - \beta) (dW_t)^2$$

Then:

$$W_t = W_{t-1} - \alpha \frac{dW_t}{\sqrt{s_t} + \epsilon}$$

So the update is:

$$\Delta W = -\alpha \frac{dW_t}{\sqrt{s_t} + \epsilon}$$

Notice that RMSProp does NOT use  $s_t$  as the direction.

It uses:

- $dW_t \rightarrow$  direction
  - $\sqrt{s_t} \rightarrow$  scale of the gradient
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## Why divide by the root of velocity?

Suppose we have two parameters:

Parameter W1: gradient = 10  
Parameter W2: gradient = 0.1

Without RMSProp:

$$\Delta W_1 = -\alpha(10)$$

$$\Delta W_2 = -\alpha(0.1)$$

W1 gets a huge update while W2 gets a tiny update.

RMSProp says:

"If a parameter has consistently large gradients, reduce its effective learning rate."

For W1, the accumulated squared gradient becomes large:

$$s \approx 100$$

Therefore:

$$\frac{10}{\sqrt{100}} = \frac{10}{10} = 1$$

For W2:

$$s \approx 0.01$$

so:

$$\frac{0.1}{\sqrt{0.01}} = \frac{0.1}{0.1} = 1$$

Now both get updates of roughly similar scale.

That's the main idea.

## So remember this distinction

Momentum asks:

"Which direction should I move, considering previous gradients?"

RMSProp asks:

"How large should my step be for this particular parameter?"

Therefore:

Momentum: smooth the direction

RMSProp: adapt the step size

And Adam combines both:

Adam = Momentum-like direction + RMSProp-like scaling

One small terminology point: in RMSProp,  $s_t$  is usually called the moving average of squared gradients, not velocity. This distinction becomes useful when you later compare Momentum, RMSProp, and Adam.